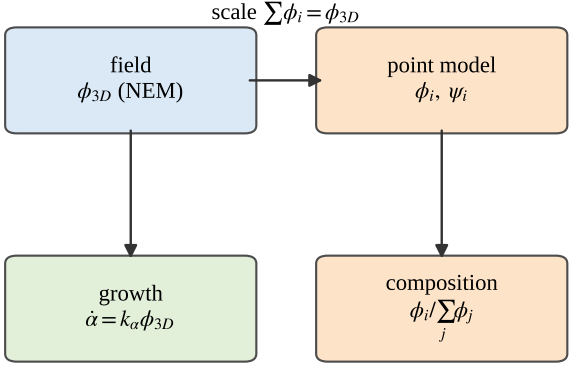
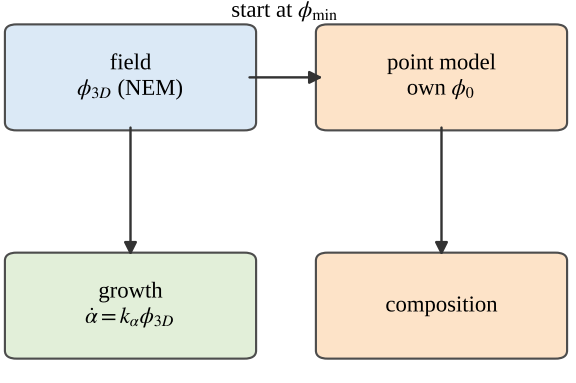
(a) growth from the amount



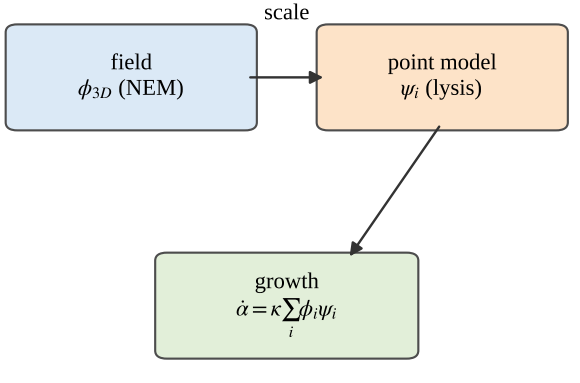
$\phi_0 = 1 - \phi_{3D}$: physical void

(b) free point model



composition depends on the local age

(c) growth from the living biomass



lower ψ_i lowers growth from inside